Graph Vertices: 0,1,2,3,4

Step 1:

Start at 0

Path: [0]

Step 2:

From 0 → try 1 ✓

Path: [0,1]

Step 3:

From 1 → try 2 ✓

Path: [0,1,2]

Step 4:

From 2 → try next vertices:

0 ✗ already visited

1 ✗ already visited

3 ✗ no edge

4 ✓ but leads to dead end later

Backtrack from 2

Step 5:

From 1 → try 3 ✓

Path: [0,1,3]

Step 6:

From 3 → try 4 ✓

Path: [0,1,3,4]

Step 7:

From 4 → try 2 ✓

Path: [0,1,3,4,2]

Step 8:

All vertices visited → check edge 2 → 0 ✓

Step 9:

Cycle completed

Final Cycle:

0 → 1 → 3 → 4 → 2 → 0